

MDSBR: Multimodal Denoising for Session-based Recommendation

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Abstract

Multimodal session-based recommendation (SBR) has emerged as a promising direction for capturing user intent using visual and textual item content. However, existing methods often overlook a fundamental issue: the modality features extracted from pre-trained models (e.g., BERT, CLIP) are inherently noisy and misaligned with user-specific preferences. This noise arises from label errors, task mismatch, and over-inclusion of irrelevant content, ultimately degrading recommendation quality. In this work, we propose a diffusion-based denoising framework that explicitly refines noisy pre-trained representations without full fine-tuning. By a structured denoising process, our Multimodal Denoising Diffusion Layer progressively eliminates the noise introduced by pre-trained models. Furthermore, we introduce two auxiliary modules: an Interest-Guided Denoising Layer that filters modality features using user interest, and a Multimodal Alignment Layer that enforces cross-modal coherence. Extensive experiments on real-world datasets demonstrate that our model significantly outperforms state-of-the-art methods while maintaining practical training efficiency. The code is available in https://github.com/YutongLi2024/MDSBR.

CCS Concepts

• **Information systems** → **Recommender systems**.

Keywords

Multimodal Recommendation; Multimodal Denoising; Pre-trained Model

ACM Reference Format:

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1 Introduction

Session-based recommendation (SBR) aims to predict users' next interaction based on their short-term behavior sequences within a session. Unlike traditional recommendation methods that rely on long-term user history, SBR is particularly useful in scenarios involving anonymous users or new users with limited historical data. Recently, multimodal SBR has gained increasing attention due to its ability to incorporate rich semantic signals such as product images, textual descriptions, and reviews, which help alleviate the sparsity and cold-start problems commonly observed in session-based settings.

However, the integration of multimodal signals is inherently accompanied by the problem of multimodal noise, which manifests in various forms and at multiple stages of the recommendation pipeline. Existing research has shown that both visual and textual content can contain substantial noise. For instance, item images may include distracting visual elements such as accessories or cluttered backgrounds, while textual descriptions often incorporate exaggerated promotional phrases or irrelevant metadata [40]. Such noise can distort item representation and hinder the model's ability to learn precise user preferences. Moreover, in collaborative contexts, user feedback itself may be noisy—due to random clicks, popularity bias, or bot activity [36, 44]. To mitigate these issues, a number of multimodal denoising recommendation models have been proposed. FREEDOM [44] freezes and sparsifies the graph structures to reduce structural noise from user feedback and modality co-occurrence. MMSBR [40] introduces pseudo-modalities and contrastive learning to reduce feature-level noise in images and text. DiffMM [12] leverages self-supervised signals for modality-aware augmentation, while DA-MRS [36] disentangles modality-specific semantic and behavioral graphs and aligns user feedback with content-driven representations. These approaches have demonstrated effectiveness in improving recommendation accuracy by addressing modality-specific noise, either through graph refinement, contrastive learning, or self-supervised filtering. Despite these advancements, we identify a fundamental yet under-explored issue in multimodal recommendation: existing methods generally assume that the representations extracted from pre-trained models are reliable and task-relevant, and treat them as fixed inputs for downstream learning. This assumption neglects the fact that pre-trained models, commonly used for feature extraction, will introduce their own noise, which can be detrimental to recommendation.

Specifically, there are two major reasons why pre-trained models can be problematic in the context of multimodal session-based recommendation: **Challenge 1: Latent noise and bias in pre-training data.** Recent studies have empirically demonstrated that even small amounts of label noise in pre-training datasets can

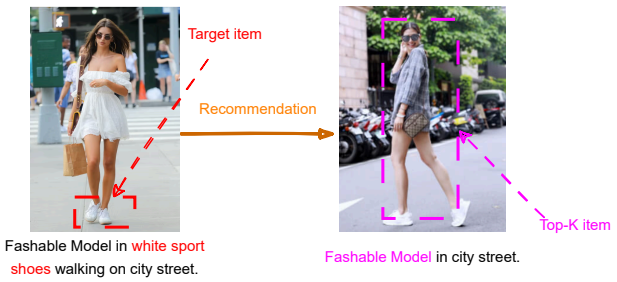


Figure 1: Pre-trained noise causes interest misalignment in recommendation. In the left image, the user focuses on the white sport shoes, but CLIP-like encoders capture all elements—dress, person, and background—diluting the true intent. As a result, the top recommendation (right image) fails to reflect the user's actual interest.

severely harm the downstream robustness and transferability of pre-trained models. For example, according to Chen et al. [2], training a ResNet-50 on ImageNet with only 5% synthetic label noise already leads to a significant degradation in performance on out-of-distribution downstream tasks. Such noise not only reduces semantic discriminability but also reshapes the feature space of the pre-trained model in harmful ways (e.g., by reducing the dominant singular values and flattening the singular value distribution). These effects result in models that overfit to noise patterns and become less generalizable. In multimodal pre-training datasets like LAION-2B [30], where noisy image-text pairs are collected from the web without strict filtering, label noise, false correlations, and domain contamination are inevitable. This form of pre-training noise propagates into downstream tasks—such as recommendation—and introduces misleading features that impair user preference modeling. **Challenge 2: Representation misalignment and over-inclusion.** Pre-trained models such as BERT [5] and CLIP [26] are typically optimized for general-purpose tasks like language modeling or image-text retrieval using corpora such as Wikipedia or LAION-400M. While effective for generic semantic representation, they often struggle to capture fine-grained, domain-specific signals critical for personalized recommendation [5, 17, 26]. As a result, they may retain irrelevant backgrounds in images or promotional phrases in text, leading to representations that overfit to irrelevant semantics. This over-inclusion of noisy or generic content, combined with a lack of task-specific discrimination, ultimately dilutes user-intent signals and hinders accurate preference modeling in recommendation tasks. Fig. 1 illustrates this issue concretely. In the left image, the user's actual interest lies in the white sport shoes; however, due to the holistic nature of CLIP-like encoders, the model encodes the entire image—including the dress, girl, and urban background—without distinguishing what truly reflects the user's intent. As a result, the top-ranked recommendation (right image) shares stylistic cues but fails to capture the user's actual target. The recommendation is semantically plausible but behaviorally misaligned. This misalignment reveals pre-trained models may encode task-irrelevant features, ultimately reducing recommendation precision. This reinforces our claim that pre-training noise is ubiquitous and must be actively addressed through a dedicated denoising mechanism.

However, adapting pre-trained encoders through full fine-tuning or training task-specific models from scratch is often impractical in real-world multimodal SBR systems. Fine-tuning requires substantial computational resources, breaks the modularity of recommender pipelines, and is infeasible when using black-box models with limited access [2, 31]. On the other hand, training new encoders demands curated multimodal datasets and high annotation cost, sacrificing scalability and cross-domain generalization, which is time-consuming and often unrealistic in realistic settings [5, 7, 17, 26]. These limitations call for a lightweight alternative that can refine noisy pre-trained features without retraining.

To address these challenges, we propose a novel paradigm to refine noisy pre-trained representations for multimodal SBR—via denoising diffusion. Unlike generative baselines like GANs [6] or VAEs [15], which often suffer from instability or oversmoothing, diffusion models provide a robust and theoretically grounded framework for structured denoising [10, 25, 27]. A diffusion model gradually corrupts input data with noise and learns a reverse process that reconstructs clean signals. This progressive denoising aligns naturally with our goal: to strip away irrelevant noise while preserving task-specific semantics within image and text embeddings [10, 12, 20]. The controllable, step-wise nature of diffusion allows us to fine-tune the denoising process at a representation level, providing flexible and adaptive refinement that is hard to achieve with conventional methods. We therefore design the Multimodal Denoising Diffusion Layer, which takes noisy modality features from pre-trained encoders and iteratively transforms them into cleaner, user-aligned representations. This module directly removes representation noise, going beyond contrastive learning by learning a task-conditioned feature refinement process.

While the diffusion process effectively removes general noise in pre-trained modality features, it operates in a modality-agnostic and user-independent manner. In session-based scenarios, where user intent is dynamic and highly contextual, this uniform denoising lacks sensitivity to changes in user interest. To address this, we introduce an interest-guided denoising mechanism that leverages short-term user behavior as a guiding signal to refine modality features. By modeling attentive interactions between session context and denoised features, the model selectively suppresses irrelevant components and preserves user-aligned semantics. Furthermore, due to inherent heterogeneity between image and text modalities, the denoised representations may still exhibit semantic discrepancies, which can impair fusion and prediction. To mitigate this, we incorporate a multimodal alignment strategy that enforces semantic consistency between the image and text embeddings of the same item. This alignment ensures coherent cross-modal representations, resulting in more accurate and context-aware recommendations. In summary, our contributions are as follows:

- We identify and formulate the problem of pre-training noise in multimodal SBR, analyze its causes, and highlight its overlooked impact on recommendation performance.
- We propose a diffusion-based denoising framework with two auxiliary modules—Interest-Guided Denoising Layer and

Multimodal Alignment Layer—that collaboratively refine noisy pre-trained features and enhance user-intent relevance and cross-modal consistency.
- Extensive experiments on real-world datasets demonstrate that our model significantly outperforms state-of-the-art methods while maintaining practical training efficiency.

2 Preliminary

2.1 Problem Statement

The multimodal SBR aims to provide personalized recommendations to anonymous users by analyzing their short-term behavioral sequences. Let $\mathcal{X} = \{x_1, x_2, ..., x_n\}$ denote the set of all items. Each item $x_i \in \mathcal{X}$ is presented to users in the form of multimodal information, including an item image x_i^{img}, description text x_i^{txt}, and item ID x_i^{id}, i.e., $x_i = \{x_i^{img}, x_i^{txt}, x_i^{id}\}$. In the multimodal SBR, a user interacts with n items in chronological order, forming a session $S = [x_1, x_2, ..., x_n]$, where $x_i \in \mathcal{X}$ is the item clicked in session S, and n is the session length. Given a session S, the goal of the system is to predict the next item x_{n+1} that the user is likely to interact with.

2.2 Diffusion Model

Before introducing our model, we briefly outline the core concepts of the standard diffusion model. This model typically involves two main processes: the forward and reverse processes [10]. In the forward process, noise is incrementally added to the data, transforming it into Gaussian noise [10]. The reverse process then reconstructs the original data by iteratively predicting and removing the added noise.

2.2.1 Forward Process. Let $\mathbf{x}_0 \sim q(\mathbf{x}_0)$ denote the original data samples. The forward process is a Markov Chain [24] that gradually adds Gaussian noise to $\mathbf{x}_0$, which can be formulated as follows:

$$\begin{aligned} q(\mathbf{x}_{1:T}|\mathbf{x}_0) &:= \prod_{t=1}^{T} q(\mathbf{x}_t|\mathbf{x}_{t-1}), \\ q(\mathbf{x}_t|\mathbf{x}_{t-1}) &:= \mathcal{N}(\mathbf{x}_t; \sqrt{1-\beta_t}\,\mathbf{x}_{t-1}, \beta_t\mathbf{I}), \end{aligned} \tag{1}$$

where T is the number of steps to add noise, $t \in \{1, 2, \ldots, T\}$ is the current step, $\beta_t \in (0, 1)$ is the noise scale at step t, $\mathbf{I}$ is the identity matrix, and $\mathcal{N}$ denotes the Gaussian distribution, meaning that $\mathbf{x}_t$ is sampled from this distribution. The values of β_t are generated from a predefined noise schedule β, which determines how much noise is injected at each step. Common noise schedules include square-root [18], cosine [10], and linear [23]. Due to the additivity of independent Gaussian noises and the reparameterization trick [10], $\mathbf{x}_t$ can be directly obtained from $\mathbf{x}_0$ using the following calculation:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\bar{\alpha}_t}\,\mathbf{x}_0, (1-\bar{\alpha}_t)\mathbf{I}), \tag{2}$$

where $\alpha_t = 1 - \beta_t$ and $\bar{\alpha}_t = \prod_{t'=1}^{t} \alpha_t$.

2.2.2 Reverse Process. The reverse diffusion process is the fundamental step in diffusion models, aiming to progressively reconstruct the original data sample x_0 from the fully perturbed sample $x_T \sim \mathcal{N}(0, I)$. This process is modeled as a Markov Chain [24], where each transition involves moving from the current state $\mathbf{x}_t$ to the preceding state $\mathbf{x}_{t-1}$. Specifically, the objective of the reverse process is to iteratively denoise the data, thereby gradually restoring the original sample. Given the current denoised representation $\mathbf{x}_t$, the subsequent representation $\mathbf{x}_{t-1}$ is computed as follows:

$$\begin{aligned} p_\theta(\mathbf{x}_{0:T}) &= p(\mathbf{x}_T) \prod_{t=1}^{T} p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t), \\ p_\theta(\mathbf{x}_{t-1}|\mathbf{x}_t) &= \mathcal{N}(\mathbf{x}_{t-1}; \mu_\theta(\mathbf{x}_t, t), \Sigma_\theta(\mathbf{x}_t, t)). \end{aligned} \tag{3}$$

Here, $\mu_\theta(\mathbf{x}_t, t)$ and $\Sigma_\theta(\mathbf{x}_t, t)$ represent the mean and covariance of the Gaussian distribution, respectively, parameterized by the neural network parameters θ. Typically, the covariance $\Sigma_\theta(\mathbf{x}_t, t)$ is set to $\sigma_t^2 I$, where σ_t^2 is a time-dependent constant calculated as:

$$\sigma_t^2 = \frac{1-\bar{\alpha}_t}{1-\bar{\alpha}_{t-1}}\beta_t \tag{4}$$

In practical applications, since the original data $\mathbf{x}_0$ is unknown, a deep neural network—such as a Multi-Layer Perceptron (MLP) [29], U-Net [28], or Transformer [33] is typically employed to estimate $\mathbf{x}_0$. This network is trained by minimizing the Variational Lower Bound (VLB) to ensure that the generated data closely approximates the true data distribution.

2.2.3 Optimization. The primary objective of the general diffusion model is to minimize the divergence between the posterior distribution in the forward process and the prior distribution in the reverse process. Consequently, the objective function can be expressed in the form of the Kullback-Leibler (KL) divergence. For the sake of simplicity, this can alternatively be formulated as [10]:

$$\mathcal{L}_{diff} = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\bar{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\bar{\alpha}_t}\epsilon, t\right)\right\|^2\right] \tag{5}$$

3 Methodology

3.1 Multimodal Information Embedding

In multimodal SBR, we encode image (x_i^{img}), text (x_i^{txt}) and item ID (x_i^{id}) to represent items comprehensively. We can obtain the image embeddings $\mathbf{e}_i^{img} \in \mathbb{R}^d$, text embeddings $\mathbf{e}_i^{txt} \in \mathbb{R}^d$ and item ID embeddings $\mathbf{e}_i^{id} \in \mathbb{R}^d$:

$$\mathbf{e}_i^{img} = imgEmb(x_i^{img}), \tag{6}$$

$$\mathbf{e}_i^{txt} = txtEmb(x_i^{txt}), \tag{7}$$

$$\mathbf{e}_i^{id} = IDEmb(x_i^{id}), \tag{8}$$

where $imgEmb(\cdot)$ and $txtEmb(\cdot)$ represent the ResNet [7] and BERT [5], respectively, without any fine-tuning. This decision ensures that the improvements of our model are attributed to the structural enhancements for denoising rather than the strength of state of the art pre-trained models. The $IDEmb(\cdot)$ represents a trainable ID embedding layer. The item ID embedding e_i^{id} is directly employed in subsequent processes without additional refinement.

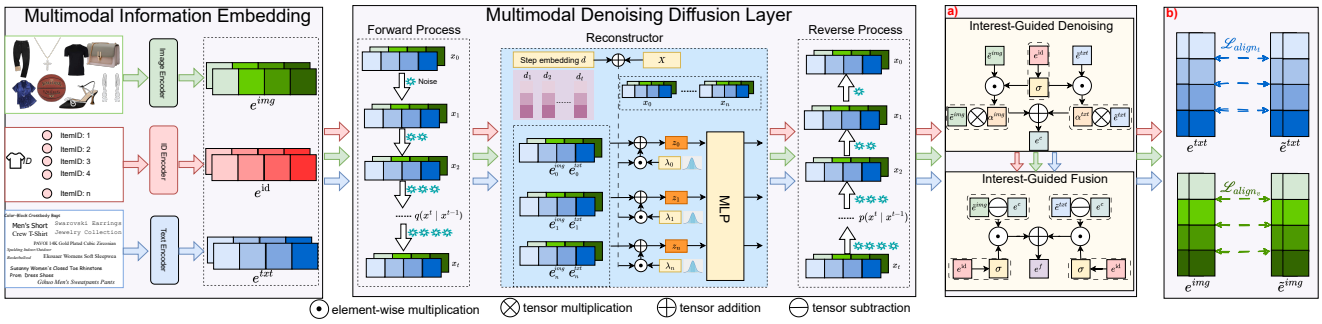


Figure 2: The Structure of MDSBR. a) The Structure of Interest-Guided Denoising Layer. b) The Structure of Multimodal Alignment Layer.

3.2 Multimodal Denoising Diffusion Layer

To address inherent noise in image and text embeddings, we propose the Multimodal Denoising Diffusion Layer. This layer refines multimodal embeddings through a diffusion process that progressively reduces noise, enhancing the clarity of representations. The process consists of **Forward Process**, which introduces noise during training, and **Reverse Process,** which denoises during inference. To enhance denoising, our model incorporates a **Reconstructor** to improve session modeling and optimize target item representations. By employing a loss function that balances noise reduction and feature preservation, the model generates high-quality multimodal representations tailored for downstream tasks.

3.2.1 Reconstructor. Based on [34, 43], we employ a MLP [29] as the core of the reconstructor function, denoted as $f_\theta(\cdot)$, to generate the reconstructed item representation $\mathbf{x}_0$ during both the forward and reverse processes. Although more complex architectures like Transformer or U-Net could be employed [20, 21], we choose MLP to simplify the computational complexity and clearly demonstrate that the performance improvements of our model stem from the denoising mechanism rather than reliance on advanced model structures. Specifically, the reconstructor maps input features $\mathbf{z}_i$ to the output representation, and the input features $\mathbf{z}_i$ are defined as:

$$\mathbf{z}_i = \mathbf{e}_i^m + \lambda_i \odot (\mathbf{x}_i + \mathbf{d}), \tag{9}$$

where $m \in \{img, txt\}$, and $\odot$ denotes element-wise multiplication. The term λ_i, sampled from a normal distribution $\lambda_i \sim \mathcal{N}(\delta, \delta)$, introduces noise, with δ controlling both the mean and variance. The vector $\mathbf{x}$ represents the target item embedding. In the forward process, it is the noisy target embedding $\mathbf{x}_s$; in the reverse process, it is the denoised target embedding as the step index s decreases from t to 1. The vector $\mathbf{d}$ is a step-specific embedding, following the approach in [10], which captures the unique characteristics of each diffusion step. The final output of the reconstructor is the reconstructed representation $\mathbf{x}_0 = f_\theta(\mathbf{z}_i)$.

3.2.2 Forward Process. Refer to Fig. 2. Following the diffusion model in Section 2, we apply the reverse process to recover the target item from the session $\mathcal{S}$. Noise is progressively added to the target item embedding $\mathbf{e}_n$ during the diffusion process, as outlined in Equation 1. The noisy representation $\mathbf{x}_t$, sampled from $q(\cdot)$, undergoes t-diffusion steps, which helps the embedding retain latent aspects of the target item. The forward process can be formally expressed as:

$$\mathbf{x}_t \leftarrow q(\mathbf{x}_t|\mathbf{x}_0, t), \tag{10}$$

where $\mathbf{x}_0$ is the reconstructed representation of reconstructor. Following Equation 10, we generate $\mathbf{x}_t$ and derive $\mathbf{x}_0$ with one-step diffusion from the target item embedding:

$$q(\mathbf{x}_t|\mathbf{x}_0) = \mathcal{N}(\mathbf{x}_t; \sqrt{\overline{\alpha}_t}\mathbf{x}_0, (1-\overline{\alpha}_t)\mathbf{I}), \tag{11}$$

where $\overline{\alpha_t} = \prod_{t'=1}^{t} \alpha_t$, and $\alpha_t = 1 - \beta_t$. Here, $\beta_t \in (0, 1)$ controls the Gaussian noise scales added at each step t. Then we use the reparameterization trick ($\epsilon \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$), we generate $\mathbf{x}_t$ as:

$$\mathbf{x}_t = \sqrt{\overline{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\overline{\alpha}_t}\epsilon \tag{12}$$

We use a linear variance noise schedule [43] to regulate the noise level:

$$1 - \overline{\alpha}_t = s \cdot \left[\alpha_{\min} + \frac{t-1}{T-1}(\alpha_{\max} - \alpha_{\min})\right], \quad t \in \{1, \ldots, T\}, \tag{13}$$

where $\alpha_{\min}$ and $\alpha_{\max} \in (0, 1)$ are the noise bounds, and $s \in (0, 1)$ controls the noise scale.

3.2.3 Reverse Process. Next, we continue to use the $\mathbf{x}_0$ from the reconstructor for estimation. Following Equation 3, the reverse step is performed after applying reparameter trick:

$$\mathbf{x}_{t-1} = \tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) + \tilde{\beta}_t\epsilon', \tag{14}$$

where $\tilde{\mu}_t(\mathbf{x}_t, \mathbf{x}_0) = \frac{\sqrt{\overline{\alpha}_{t-1}}\beta_t}{1-\overline{\alpha}_t}\mathbf{x}_0 + \frac{\sqrt{\alpha_t}(1-\overline{\alpha}_{t-1})}{1-\overline{\alpha}_t}\mathbf{x}_t$ and $\tilde{\beta}_t = \frac{1-\overline{\alpha}_{t-1}}{1-\overline{\alpha}_t}\beta_t$, and $\epsilon' \sim \mathcal{N}(\mathbf{0}, \mathbf{I})$ for $\mathbf{x}_{t-1}$ generation. Repeat the above process until we arrive at $\mathbf{x}_1$, and then estimate $\mathbf{x}_0$.

3.2.4 Diffusion Loss Function. As the reconstructor requires the step embedding for reconstruction according to Equation 3.2.1, in Forward Process, we randomly sample a step-index s from a uniform distribution over range $[1, t]$. So as to the Reverse Process, the step-index will be from t to 1, where t is the total steps. Therefore, the corresponding step embedding and sequence item distribution are fed into the reconstructor for model learning. According to Equation 5, The standard objective function of diffusion models could be simplified as a mean-squared error:

$$\mathcal{L}_{diff} = \mathbb{E}_{t,\mathbf{x}_0,\epsilon}\left[\left\|\epsilon - \epsilon_\theta\left(\sqrt{\overline{\alpha}_t}\mathbf{x}_0 + \sqrt{1-\overline{\alpha}_t}\,\epsilon, t\right)\right\|^2\right]. \tag{15}$$

Finally, $\tilde{\mathbf{e}}_i^{img}$ and $\tilde{\mathbf{e}}_i^{txt}$ denote the image and text embeddings after being processed by the Multimodal Diffusion Denoising Layer.

3.3 Interest-Guided Denoising Layer

In SBR, irrelevant modality information can obscure key signals and reduce recommendation accuracy. Such noise includes unnecessary visual or textual details that do not reflect user preferences. To address this, we utilize user behavior as a guiding signal to filter and refine modality features, ensuring the retained features align closely with user interests.

3.3.1 Interest-Guided Denoising. Multimodal data, such as product images and textual descriptions, often contain redundant or irrelevant content that does not align with a user's true preferences. For instance, a product image may contain complex background elements, and a description might include generic marketing terms. These noisy elements can dilute the recommendation model's ability to capture what the user truly likes. To address this, we propose an *Interest-Guided Denoising* mechanism that adaptively filters modality features using signals derived from user interactions.

Our key motivation lies in the observation that user-item interaction histories—particularly item IDs—encode implicit user interests. While item IDs themselves are symbolic, their embedding vectors $\mathbf{e}^{id}$, trained over large-scale behavioral data, can serve as robust interest priors. We leverage these embeddings to guide the denoising of both visual and textual modalities, aligning them with user preferences and discarding content that lacks relevance. To achieve this, we introduce a multi-layer interest extractor $f(\cdot)$ that transforms $\mathbf{e}^{id}$ into an informative attention signal. Specifically, we adopt a two-layer perceptron with non-linear activation and residual enhancement:

$$\mathbf{h}_1 = \sigma(\mathbf{W}_1\mathbf{e}^{id} + \mathbf{b}_1), \tag{16}$$

$$\mathbf{h}_2 = \sigma(\mathbf{W}_2\mathbf{h}_1 + \mathbf{b}_2), \tag{17}$$

$$f(\mathbf{e}^{id}) = \mathbf{h}_2 + \mathbf{e}^{id}, \tag{18}$$

where σ is a non-linear activation function (e.g., ReLU or GELU), and $\mathbf{W}_1, \mathbf{W}_2 \in \mathbb{R}^{d\times d}$, $\mathbf{b}_1, \mathbf{b}_2 \in \mathbb{R}^d$ are learnable parameters. The residual connection ensures that the user interest signal retains the original behavior information while enabling more expressive transformations. We then apply this learned interest representation $f(\mathbf{e}^{id})$ to guide modality-specific attention over the denoised multimodal embeddings $\tilde{\mathbf{e}}^{img}$ and $\tilde{\mathbf{e}}^{txt}$, which are preprocessed via a denoising diffusion model to remove low-level and distributional noise. The attention scores are computed as:

$$\mathbf{a}^m = \text{softmax}(\tilde{\mathbf{e}}^m \odot f(\mathbf{e}^{id})), \quad m \in \{img, txt\}, \tag{19}$$

where $\odot$ denotes element-wise multiplication. These scores highlight modality components most aligned with user interests. Based on these attentions, we compute a common representation $\mathbf{e}^c$ that captures shared user-relevant information across modalities:

$$\mathbf{e}^c = \mathbf{a}^{img}\tilde{\mathbf{e}}^{img} + \mathbf{a}^{txt}\tilde{\mathbf{e}}^{txt}. \tag{20}$$

This representation effectively filters out cross-modal noise and extracts commonalities that reflect consistent user preferences.

3.3.2 Interest-Guided Fusion. To preserve modality-specific nuances while avoiding redundancy, we further extract unique features from each modality by removing the shared representation $\mathbf{e}^c$. These residual features are then reweighted by the interest signal to emphasize their relevance:

$$\mathbf{e}^f = (\tilde{\mathbf{e}}^{img} - \mathbf{e}^c) \odot f(\mathbf{e}^{id}) + (\tilde{\mathbf{e}}^{txt} - \mathbf{e}^c) \odot f(\mathbf{e}^{id}), \tag{21}$$

where $\mathbf{e}^f$ captures complementary and non-overlapping user-aware information from both modalities. Finally, we construct the full item representation by concatenating the common representation, the unique features, and the original interest embedding:

$$y_i = \mathbf{e}^f \parallel \mathbf{e}^c \parallel \mathbf{e}^{id}, \tag{22}$$

where $\parallel$ denotes concatenation and $y_i \in \mathbb{R}^{3d}$. This fused representation effectively encodes personalized signals while eliminating noise, leading to enhanced recommendation accuracy.

3.4 Multimodal Alignment Layer

While the Interest-Guided Denoising module effectively suppresses modality-irrelevant noise guided by user interests, it primarily focuses on intra-modality filtering. However, in practice, the generated denoised embeddings $\tilde{\mathbf{e}}_i^{img}$ and $\tilde{\mathbf{e}}_i^{txt}$, obtained through the diffusion process, may still exhibit cross-modal discrepancies due to semantic inconsistencies between image and text. Such inconsistencies can degrade the effectiveness of subsequent fusion and prediction stages.

To address this issue, we introduce a *Multimodal Alignment Layer* that bridges the semantic gap between modalities by explicitly aligning the generated representations $\tilde{\mathbf{e}}_i^{img}, \tilde{\mathbf{e}}_i^{txt}$ with their original counterparts $\mathbf{e}_i^{img}, \mathbf{e}_i^{txt}$. The motivation is twofold: first, to preserve the rich semantic information inherently encoded in the original modalities; second, to guide the denoised outputs toward semantically meaningful and structurally consistent representations. This design complements the Interest-Guided Denoising module by refining inter-modal coherence after intra-modal noise suppression. Inspired by prior work [22], we model both the original and denoised modality embeddings as Gaussian distributions, encouraging statistical consistency across means and variances. This probabilistic formulation enables the model to reduce representational drift introduced during the diffusion-based denoising stage.

$$\tilde{\mathbf{e}}_i^{img} \sim \mathcal{N}(\mu_{\tilde{img}}, \sigma^2_{\tilde{img}}), \quad \mathbf{e}_i^{img} \sim \mathcal{N}(\mu_{img}, \sigma^2_{img}) \tag{23}$$

$$\tilde{\mathbf{e}}_i^{txt} \sim \mathcal{N}(\mu_{\tilde{txt}}, \sigma^2_{\tilde{txt}}), \quad \mathbf{e}_i^{txt} \sim \mathcal{N}(\mu_{txt}, \sigma^2_{txt}) \tag{24}$$

Here, $(\mu_{\tilde{img}}, \sigma_{\tilde{img}})$ and $(\mu_{\tilde{txt}}, \sigma_{\tilde{txt}})$ are the parameters learned from the denoised representations, while $(\mu_{img}, \sigma_{img})$ and $(\mu_{txt}, \sigma_{txt})$ are derived from the original modality encoders. The alignment loss penalizes distributional mismatches between the generated and original modalities, encouraging semantic convergence:

$$\mathcal{L}_{align_v} = \left\|\mu_{\tilde{img}} - \mu_{img}\right\|_1 + \left\|\sigma_{\tilde{img}} - \sigma_{img}\right\|_1 \tag{25}$$

$$\mathcal{L}_{align_t} = \left\|\mu_{\tilde{txt}} - \mu_{txt}\right\|_1 + \left\|\sigma_{\tilde{txt}} - \sigma_{txt}\right\|_1 \tag{26}$$

$$\mathcal{L}_{align} = \mathcal{L}_{align_v} + \mathcal{L}_{align_t} \tag{27}$$

By enforcing consistency in distributional parameters, this alignment step ensures that denoised modalities remain semantically

faithful to their original inputs, thereby enhancing the reliability and interpretability of downstream multimodal fusion. Working in tandem with Interest-Guided Denoising, this layer ensures both intra-modal noise suppression and inter-modal consistency, yielding a more robust and semantically coherent recommendation framework.

3.5 Model Prediction

To optimize our model, we adopt the standard cross-entropy loss function, which is widely used in both academic literature and industrial systems [38, 39]. The recommendation task loss is defined as

$$\mathcal{L}_{rec} = -\sum_{i=1} \left(y_i \log(\hat{y}_i) + (1 - y_i) \log(1 - \hat{y}_i)\right). \tag{28}$$

where y_i denotes the ground truth label (one-hot vector), and $\hat{y}_i$ represents the predicted probability. Finally, we combine the recommendation objective with the alignment and diffusion losses. The total loss for the model is defined as

$$\mathcal{L} = \mathcal{L}_{rec} + \alpha \cdot \mathcal{L}_{align} + \beta \cdot \mathcal{L}_{diff}, \tag{29}$$

where $\mathcal{L}_{rec}$ captures the recommendation objective, $\mathcal{L}_{align}$ enforces cross-modal consistency, and $\mathcal{L}_{diff}$ improves representation denoising via diffusion modeling.

4 Experiments

In this section, we study the following research questions: **RQ1:** How does MDSBR perform compared to the baseline methods? **RQ2:** How do different components affect MDSBR? **RQ3:** What influence do different hyperparameters have on the results? **RQ4:** Can our model denoising provide explainability? **RQ5:** How efficient is our MDSBR?

4.1 Experimental Settings

4.1.1 Datasets. We evaluate our model on four publicly available multimodal datasets: Amazon-Cellphones, Amazon-Grocery, Amazon-Sports, and Yelp. The Amazon subsets are from the 2018 release of the Amazon product dataset,[1] and the Yelp dataset is collected from its official open data portal.[2] Each dataset contains item images and textual metadata. Following standard SBR protocols [40–42], we construct sessions by grouping interactions from the same user within a single day, treating each session as an independent sequence without user identity. We filter out sessions with fewer than two items and items occurring less than five times. Data is split chronologically into training, validation, and test sets using a 7:2:1 ratio. Dataset statistics are summarized in Table 1.

Table 1: Statistics of all datasets.

Datasets	Cellphones	Grocery	Sports	Yelp
#item	8,614	11,638	18,796	287,116
#interaction	196,376	364,728	566,504	148,523
#session	78,026	127,548	211,959	4,392,169
avg.length	2.52	2.86	2.67	15.3

[1] https://cseweb.ucsd.edu/~jmcauley/datasets/amazon_v2
[2] https://business.yelp.com/data/resources/open-dataset

4.1.2 Evaluation Protocols. We adopt two standard metrics: Prec@K (Precision) and MRR@K (Mean Reciprocal Rank), with K ∈ {10, 20}, following [19, 35, 40, 41]. Higher values indicate better performance.

4.1.3 Baseline Methods. We compare our model with strong baselines from three categories: (1) **ID-based models:** These rely on item co-occurrence patterns, including SASRec [14], Bert4Rec [32], LightGCN [9], DiffuRec [20], and DiffuASR [21]; (2) **Multimodal models:** These incorporate additional modalities such as item images or texts, including VBPR [8], UniSRec [11], CoHHN [42], and DIMO [41]; (3) **Multimodal denoising models:** These address noise in multimodal signals, including FREEDOM [44], MMSBR [40], and DiffMM [12].

4.1.4 Implementation Details. For each baseline, we either adopt the optimal hyperparameters reported in the original papers or conduct a grid search when the dataset was not covered in prior work, ensuring fair comparisons. For our proposed model, all hyperparameters are tuned on the validation set using grid search. We focus primarily on two key hyperparameters: the modality alignment weight α and the diffusion loss weight β, which control the strength of alignment and denoising components, respectively. We search α in {0.01, 0.05, 0.1, 0.5, 1.0, 2.0} and β in {1e-4, 1e-3, 1e-2, 1e-1, 1e-0}. The final selected values for each dataset are reported in Section 4.4. All experiments are repeated five times with different random seeds, and average results are reported. Statistical significance is assessed using paired t-tests with $p < 0.05$.

4.2 Overall Performance Comparison (RQ1)

Table 2 reports the performance of MDSBR and various baselines across four datasets, leading to the following observations: (1) **ID-based methods** (e.g., SASRec, Bert4Rec, LightGCN) generally underperform due to sparse session data and limited contextual cues. For example, LightGCN obtains Prec@20 scores of 26.18 and 34.99 on the Cellphones and Yelp datasets, respectively—substantially lower than those of MDSBR. By leveraging multimodal information, MDSBR enriches contextual understanding and better captures user preferences, effectively addressing the limitations of ID-based models. (2) **Multimodal methods** (e.g., VBPR, UniSRec, DIMO) utilize image and text features to enhance accuracy but lack mechanisms to mitigate noise in multimodal data, such as irrelevant visual content or textual bias. Consequently, their performance remains limited in noisy settings. For instance, DIMO achieves Prec@20 scores of 27.73 and 36.92 on the Cellphones and Yelp datasets, whereas MDSBR achieves 30.35 and 39.66, respectively—demonstrating the benefits of its comprehensive denoising strategy. (3) **Multimodal denoising methods** (e.g., FREEDOM, MMSBR, DiffMM) aim to filter noisy modality signals but often overlook inherent noise in pre-trained models. In contrast, MDSBR integrates three components—Multimodal Denoising Diffusion Layer, Interest-Guided Denoising Layer, and Multimodal Alignment Layer—to jointly remove irrelevant signals, align modalities, and enhance personalized representations. For example, on the Grocery dataset, MDSBR achieves Prec@20 = 45.44, outperforming MMSBR (44.21) and DiffMM (43.99). Overall, MDSBR consistently outperforms ID-based, multimodal,

Table 2: Overall Performance Comparison Table. The best and second-best results are bold and underlined, respectively. All results are averaged over 5 runs per dataset to ensure statistical robustness and are statistically significant with $p < 0.05$.

Datasets	Metrics	ID-based Methods					Multimodal Methods				Multimodal Denoising Methods			MDSBR
		SASRec (ICDM'18)	Bert4Rec (CIKM'19)	LightGCN (SIGIR'20)	DiffuRec (TOIS'23)	DiffuASR (CIKM'23)	VBPR (AAAI'16)	UniSRec (KDD'20)	CoHHN (SIGIR'22)	DIMO (SIGIR'24)	FREEDOM (MM'23)	MMSBR (TKDE'23)	DiffMM (MM'24)	
Cellphones	Prec@10	21.13	21.31	22.89	22.68	22.92	20.74	22.73	23.60	23.73	23.60	23.37	24.31	**25.17**
	Prec@20	24.12	24.16	26.18	26.33	24.51	23.21	25.65	27.71	27.73	27.60	27.22	28.84	**30.35**
	MRR@10	12.42	12.51	13.61	13.82	14.53	12.29	14.36	15.37	14.73	15.60	15.77	15.47	**16.38**
	MRR@20	13.66	13.89	14.78	14.63	14.77	13.44	14.63	14.73	15.76	15.60	15.71	15.61	**16.49**
Grocery	Prec@10	39.35	39.94	40.82	41.28	41.34	38.92	40.40	41.48	41.82	41.62	41.19	42.37	**43.54**
	Prec@20	41.41	41.02	42.74	43.24	43.40	40.79	42.44	43.59	41.58	42.10	44.21	43.99	**45.44**
	MRR@10	33.53	33.76	34.89	34.60	35.25	33.06	34.12	35.33	35.38	35.44	35.51	35.67	**36.47**
	MRR@20	33.62	33.92	35.01	34.75	35.47	33.21	34.24	35.58	35.49	35.67	35.94	35.76	**36.58**
Sports	Prec@10	30.99	31.09	32.36	32.26	32.26	30.61	31.92	32.33	32.11	32.14	32.29	32.01	**33.68**
	Prec@20	33.17	33.11	34.29	34.55	34.75	32.76	34.41	35.12	35.77	35.55	35.74	35.61	**36.23**
	MRR@10	25.95	25.78	26.45	27.17	27.54	24.22	26.21	27.13	28.11	28.32	28.15	27.69	**28.76**
	MRR@20	26.13	26.27	26.41	27.46	27.61	25.29	26.64	27.41	28.31	28.61	28.29	27.99	**29.14**
Yelp	Prec@10	33.52	33.43	34.65	34.94	35.19	33.12	34.43	36.14	35.43	36.14	36.21	36.12	**37.63**
	Prec@20	33.93	33.86	34.99	35.28	35.79	33.61	34.91	36.86	36.92	37.24	37.33	38.03	**39.66**
	MRR@10	25.14	25.32	26.48	27.21	27.67	24.13	26.54	29.56	30.14	30.06	30.31	30.12	**32.74**
	MRR@20	25.45	25.68	27.10	27.87	27.75	25.12	26.21	30.15	31.27	30.21	31.14	31.15	**32.87**

and denoising baselines across all datasets, validating its robustness and effectiveness in handling noisy multimodal session-based recommendation.

4.3 Ablation Study (RQ2)

To assess the contribution of each module in MDSBR to multimodal noise reduction and overall performance, we conduct an ablation study by progressively removing specific components. Table 3 reports the results on four datasets using Prec@10 and MRR@10. We compare the full model with the following variants: **(A) MDSBR**: Our Model with all components. **(B)** w/o **DF**: Without the Multimodal Denoising Diffusion Layer. **(C)** w/o **GD**: Without the Interest-Guided Denoising module. **(D)** w/o **GF**: Without the Interest-Guided Fusion mechanism. **(E)** w/o **MA**: Without the Multimodal Alignment Layer. **(F) Base Model**: All denoising and alignment components removed.

(A) MDSBR achieves the best overall performance across all datasets, confirming the effectiveness of our full model design. Compared to the base model (F), it brings consistent improvements in both Prec@10 and MRR@10, demonstrating the necessity of comprehensive denoising for multimodal SBR. **(B) w/o DF** Removing the diffusion layer leads to the largest performance drop. For example, on the Cellphones dataset, Prec@10 falls from 25.17 to 23.34 and MRR@10 from 16.38 to 14.34. This highlights the importance of iterative denoising in removing harmful modality noise **(C) w/o GD** Excluding the Interest-Guided Denoising module, which filters modality-specific noise via user interest signals, results in consistent degradation. On Yelp, for instance, Prec@10 drops from 37.63 to 36.25, confirming the value of user-aware denoising. **(D) w/o GF** Without guided fusion, the model struggles to effectively integrate denoised features. Though each modality retains useful information, improper fusion can reintroduce noise or weaken signals. On Sports, MRR@10 drops from 28.76 to 28.44, illustrating the necessity of interest-guided fusion. **(E) w/o MA** Removing the alignment layer compromises cross-modal consistency. Misaligned modality distributions hinder the model's ability to exploit complementary information. On the Grocery dataset, Prec@10 decreases from 43.54 to 42.24. **(F) Base Model** The base model, lacking all denoising and alignment modules, performs worst across all datasets. This underscores the effectiveness of each proposed component in reducing noise and enhancing representation quality.

Table 3: The Prec@10 and MRR@10 performances achieved by MDSBR variants on four datasets. All results are averaged over 5 runs per dataset to ensure statistical robustness and are statistically significant with $p < 0.05$.

Model	Cellphones		Grocery		Sports		Yelp	
	Prec@10	MRR@10	Prec@10	MRR@10	Prec@10	MRR@10	Prec@10	MRR@10
(A) MDSBR	**25.17**	**16.38**	**43.54**	**36.47**	**33.68**	**28.76**	**37.63**	**32.74**
(B) w/o *DF*	23.34	14.34	42.18	34.11	32.93	27.01	35.68	30.24
(C) w/o *GD*	24.46	15.68	43.92	35.46	33.46	28.26	36.25	31.62
(D) w/o *GF*	24.66	15.84	43.76	35.58	33.36	28.44	36.15	31.57
(E) w/o *MA*	23.38	14.41	42.24	34.28	33.01	27.18	35.75	30.57
(F) Base Model	21.18	12.33	39.11	32.62	29.47	24.71	33.92	28.05

4.4 Hyperparameter Analysis (RQ3)

4.4.1 Hyperparameter α (Modality Alignment Balance). As shown in Fig. 3, α controls the weight of the alignment loss $\mathcal{L}_{align}$, which reduces inconsistencies between generated and original modality representations. Moderate values (e.g., $\alpha = 0.5$) yield the best performance on datasets such as Cellphones, Grocery, and Sports, striking a balance between enforcing alignment and preserving user-specific features. Low values ($\alpha < 0.1$) fail to suppress cross-modal noise, while high values ($\alpha > 0.9$) tend to overfit to alignment, reducing personalization and degrading performance, particularly in datasets with diverse user behavior. Notably, datasets with complex modality interactions, like Yelp, benefit from higher α, indicating the need for dataset-specific tuning.

4.4.2 Hyperparameter β (Diffusion Loss Weight). The hyperparameter β controls the influence of the diffusion loss $\mathcal{L}_{diff}$, which governs the denoising trajectory of pre-trained modality representations. As illustrated in Fig. 3, we observe a consistent trend across all datasets: increasing β from 1e-4 to 1e-1 leads to improved performance, indicating that a moderate diffusion strength enhances the model's ability to remove noise and capture user-relevant semantics. However, further increasing β to 1e-0 results in a performance drop,

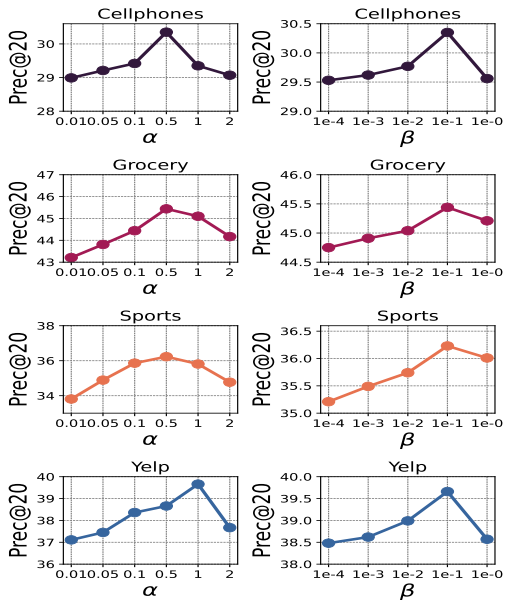


Figure 3: Performance of MDSBR with different values of α and β

suggesting that excessive reliance on the diffusion objective may distort useful information or hinder task-specific learning. These results imply that diffusion should play an important but balanced role strong enough to enable effective denoising, yet not so dominant that it overshadows collaborative or behavioral signals. Proper tuning of β is therefore essential to ensure an optimal trade-off between denoising effectiveness and information preservation.

4.5 Case Study (RQ4)

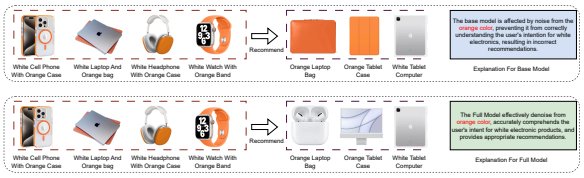


Figure 4: Case study for MDSBR. The Base Model is MDSBR with all denoising components removed. The Full Model is MDSBR.

As shown in Fig. 4, the Base Model struggles to remove irrelevant noise, limiting its ability to distinguish important features from background elements in complex images. For instance, it may mistakenly recommend products based on visual prominence—such as bright colors like orange—rather than the user's actual preference for white electronics. This misinterpretation leads to recommendations misaligned with user intent regarding product type or color. In contrast, the Full Model (MDSBR) effectively filters out such noise, focusing on the item of interest while ignoring distracting background details. It accurately identifies user-preferred attributes, such as the color white, without being influenced by irrelevant visual cues. This improved noise suppression enables more accurate and personalized recommendations that better reflect the user's true preferences.

4.6 Efficiency Analysis (RQ5)

The overall computational complexity of MDSBR is dominated by the Multimodal Denoising Diffusion Layer, which adopts a diffusion framework similar to DiffuRec [20]. This layer incurs a time complexity of $O(nd^2 + nd)$, where n is the number of samples and d is the feature dimension, with the dominant term being $O(nd^2)$. The Interest-Guided Denoising Layer involves matrix-vector operations for interest extraction ($O(nd^2)$), attention computation ($O(nd)$), and modality fusion ($O(nd)$), resulting in an overall complexity of $O(nd^2)$. The Multimodal Alignment Layer primarily computes visual-textual alignment through mean, variance, and contrastive objectives, each with complexity $O(nd)$. Therefore, the total time complexity of MDSBR remains $O(nd^2)$, indicating that the cost grows linearly with the number of samples and quadratically with feature dimension.

Table 4: Average per-epoch inference time (seconds) and Prec@10 of denoising variants across four datasets. The Base Model is MDSBR without denoising. Results are averaged over 5 runs and are statistically significant ($p < 0.05$).

Datesets	Base Model		GAN		VAE		MDSBR	
	Prec@10	time	Prec@10	time	Prec@10	time	Prec@10	time
Cellphones	21.18	7.09s	21.11	15.23s	21.17	14.56s	25.17	18.50s
Grocery	39.11	7.75s	39.07	16.41s	39.05	15.66s	43.54	20.39s
Sports	29.47	13.93s	29.42	24.59s	29.45	27.12s	33.68	32.92s
Yelp	33.92	26.92s	33.89	67.42s	33.91	72.86s	37.63	85.67s

To evaluate the practical efficiency of our design, we compare MDSBR with two simplified alternatives that replace the diffusion component with lightweight denoising strategies: a VAE-based variant and a GAN-based variant. We also include a base model with all denoising modules removed. As shown in Table 4, MDSBR consistently achieves the highest performance in terms of Prec@10 across all datasets. For example, on the Cellphones dataset, MDSBR improves Prec@10 to 25.17, compared to 21.18 (Base Model), 21.11 (GAN), and 21.17 (VAE). Similar improvements are observed on Grocery (43.54 vs. 39.11 vs 39.07 vs 39.05), Sports (33.68 vs. 29.47 vs 29.42 vs 29.45), and Yelp (37.63 vs. 33.92 vs 33.89 vs 33.91), demonstrating that simple denoising methods with other generative models fail to capture the subtle and heterogeneous noise patterns that diffusion models can effectively address. While diffusion introduces additional inference cost—for instance, per-epoch training time reaches 32.92s on Sports and 85.67s on Yelp the resulting performance gains justify this overhead. These results support two conclusions: (1) Diffusion-based denoising provides significant improvements in recommendation quality over VAE and GAN alternatives, particularly in complex multimodal environments; and (2) the increased computational cost remains acceptable for practical use, validating the adoption of diffusion in our framework.

4.7 Depth Analysis

4.7.1 Effect of Pre-trained Encoders. We evaluate the impact of different pre-trained vision-language encoders by comparing performance across six model variants. Table 5 reports the Prec@10 results on all four datasets. Within the BaseModel group (i.e., MDSBR without denoising components), replacing ResNet and BERT with CLIP or BLIP2 leads to performance gains. For instance, on Cellphones, BaseModel(CLIP) improves from 21.18 to 22.24. Similar trends hold for other datasets, suggesting that stronger encoders improve representation quality. In the full MDSBR model, adding denoising further boosts performance. MDSBR outperforms its corresponding BaseModel variant under all encoder settings. Notably, even MDSBR with ResNet/BERT surpasses BaseModel(BLIP2) on Sports (33.68 vs. 30.17). Combining MDSBR with CLIP or BLIP2 yields the highest Prec@10. For example, MDSBR(BLIP2) achieves top scores on Grocery (44.27) and Yelp (39.09), while MDSBR(CLIP) performs best on Cellphones (28.98). These bolded values in Table 5 confirm that MDSBR complements stronger encoders effectively, while remaining competitive even with simpler ones.

Table 5: Prec@10 comparison across four datasets. The Base Model denotes MDSBR without denoising modules. All results are averaged over 5 runs and are statistically significant ($p < 0.05$).

Dataset	Base Model			MDSBR		
	BERT&ResNet	CLIP	BLIP2	BERT&ResNet	CLIP	BLIP2
Cellphones	21.18	22.24	22.05	25.17	**28.98**	28.93
Grocery	39.11	39.43	39.59	43.54	43.91	**44.27**
Sports	29.47	30.02	30.17	33.68	35.71	**35.94**
Yelp	33.92	35.32	35.41	37.63	39.04	**39.09**

4.7.2 Effect of Reconstructor Architecture. To examine the impact of the reconstructor design in MDSBR, we compare three architectures: MLP, Transformer, and U-Net. Table 6 reports their Prec@10 and per-epoch inference time across four datasets. Using Transformer or U-Net improves Prec@10 over the default MLP-based reconstructor. For example, on Cellphones, Prec@10 increases from 25.17 with MLP to **26.69** with Transformer. Similar improvements are observed on other datasets. These gains, however, come with increased inference time. On Cellphones, time per epoch rises from 18.50s (MLP) to 36.75s (Transformer). On Yelp, Transformer reaches the highest latency at 132.58s, compared to 85.67s (MLP) and 121.42s (U-Net). Overall, deeper architectures improve performance but incur greater cost. MLP offers a better trade-off between accuracy and efficiency, aligning with our design goal of attributing performance to the denoising mechanism rather than model complexity.

Table 6: Prec@10 and per-epoch inference time (seconds) of different reconstructors. MDSBR uses an MLP by default, with Transformer and U-Net as substitutes. Results are averaged over 5 runs and are statistically significant ($p < 0.05$).

Dataset	MDSBR (MLP)		MDSBR (Transformer)		MDSBR (U-Net)	
	Prec@10	Time	Prec@10	Time	Prec@10	Time
Cellphones	25.17	18.50s	**26.69**	36.75s	26.30	30.21s
Grocery	43.54	20.39s	**44.32**	42.87s	43.88	35.72s
Sports	33.68	32.92s	**35.10**	60.11s	34.61	53.36s
Yelp	37.63	85.67s	**38.91**	132.58s	38.43	121.42s

5 Related Work

5.1 Multimodal Recommendation.

To address the limitations of ID-based methods in recognizing more nuanced user preferences, multimodal approaches have emerged, incorporating additional modalities such as textual descriptions, categories, and images to enhance user intent modeling. These multimodal methods offer richer semantic information, which is particularly useful in characterizing item features and user interests. For instance, description text has been widely used to convey user preferences [11, 16, 42], while categories reflect the hierarchical nature of users' decision-making processes [3, 16]. Additionally, certain methods take into account price sensitivity to better model user behavior in specific contexts [37, 40]. Recent research has explored the integration of both image and textual data to create more granular user representations. Some pioneering works [40, 41] model the interactions between these modalities and item IDs to portray user interests at a more detailed level. However, these methods do not address noise from external pre-trained models and redundant information in multimodal fusion. This motivates us to propose the MDSBR.

5.2 Diffusion Model for Recommendation.

The diffusion model has attracted much attention recently, because of its superior performance on image generation. Diffusion model recently have shown the capability of high-quality generation [4, 25], covering conditional generation [10, 27] and unconditional generation [1, 23]. DDPM [10] propose the general process of diffusion model. Some approaches, as explored by [34], model the diffusion process by capturing the distribution of interaction probabilities, while others, like DiffuRec [20] employs diffusion models to model latent aspects of items and multi-level interests of users. Notably, DiffMM [12] enhances modality awareness with a conditional denoising diffusion model for reconstructing user-item graphs and self-supervision via cross-modal data augmentation. Furthermore, DiffuASR [21] presents a sequential U-Net architecture that adapts the diffusion noise prediction model for discrete sequence generation tasks. DiffKG [13] enables robust knowledge graph learning with diffusion models.

6 Conclusion

In this paper, we present the Multimodal Denoising Diffusion for Session-based Recommendations (MDSBR). This model effectively addresses noise from pre-train model thereby enhancing the overall performance and accuracy of the recommendation system. Extensive experiments conducted on four real-world benchmark datasets have validated the effectiveness of our proposed model. However, our approach has certain limitations. Currently, the multimodal representations are not trained end-to-end, which restricts the model's flexibility and adaptability to unseen data. Nevertheless, our paper offers a promising direction for future research on leveraging multimodal information for session-based recommendation. We hope that MDSBR will inspire further exploration.

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